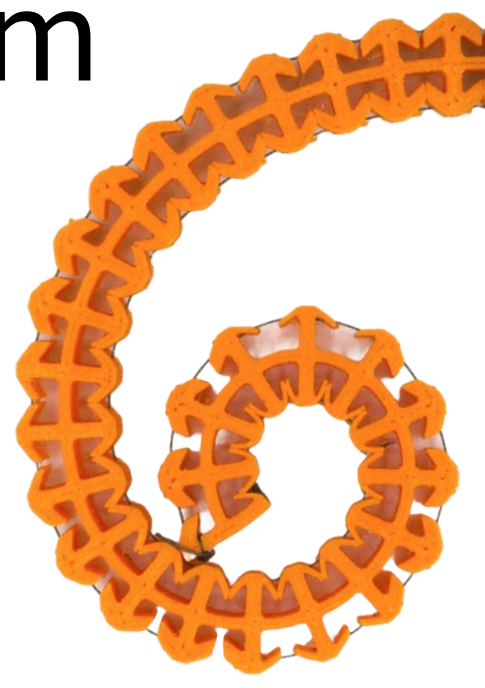


# Tendon-Driven Soft Grasper with Asymmetric Actuation through a Jamming Mechanism

**Paola Romero, Qifan Yu, Kaitlyn Becker**

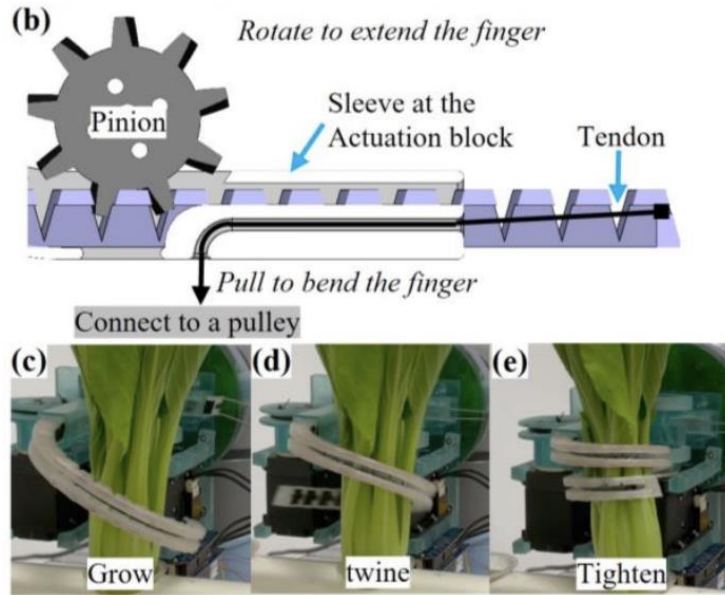




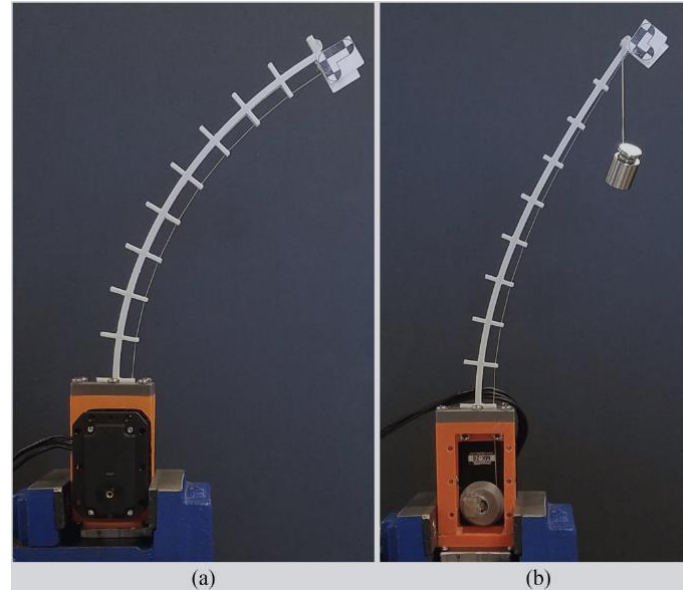
# Paola Romero

- Rising senior, undergraduate studying mechanical engineering at MIT
- Last year, developed design of soft graspers in the Fabrication Integrated Design Lab (FIDL)

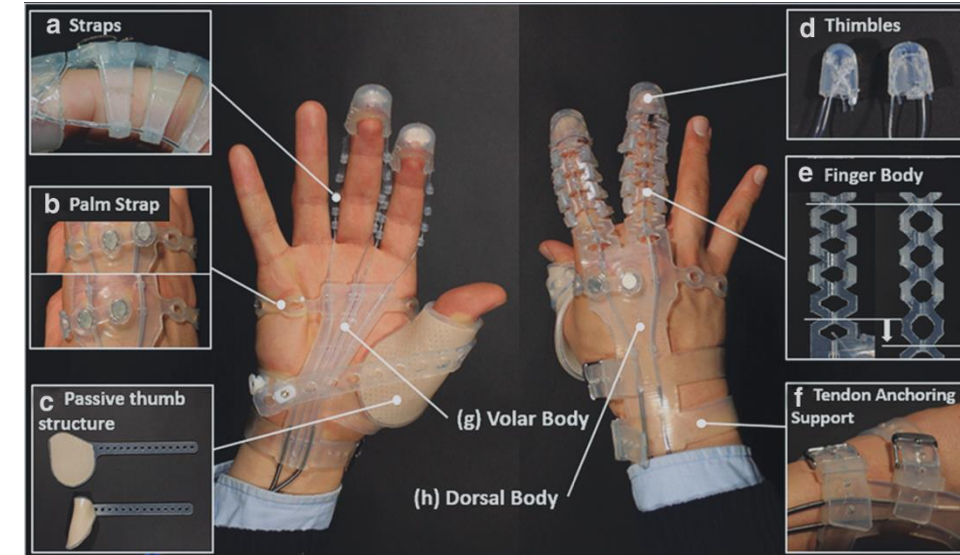
# Tendon-Driven Soft Graspers



[1]



[3]



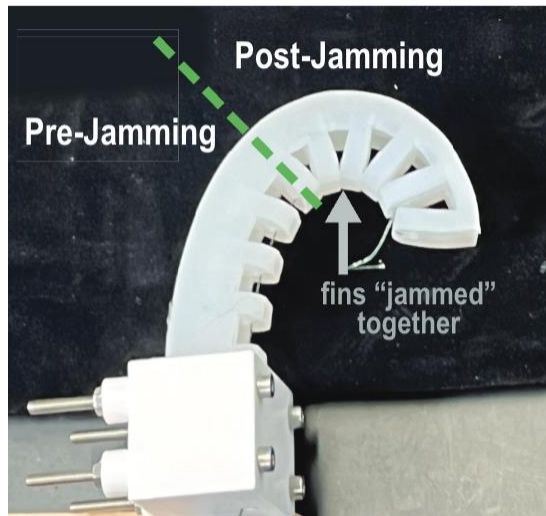
[2]

[1] Liu, J., Lai, W., Sim, B. R., Rui Tan, J. M., Magdassi, S., and Phee, S. J., 2024, "Smart Grow-and-Twine Gripper for Vegetable Harvesting in Vertical Farms," *7th IEEE International Conference on Soft Robotics, RoboSoft 2024, April 14, 2024 - April 17, 2024*, Institute of Electrical and Electronics Engineers Inc., San Diego, CA, United states, pp. 460–466. <https://doi.org/10.1109/RoboSoft60065.2024.10521949>.

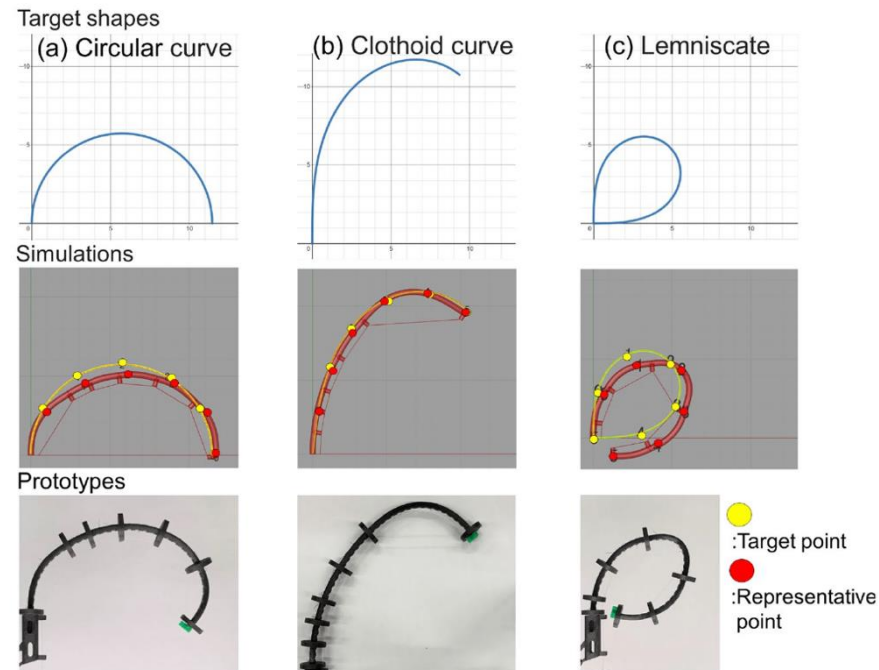
[2] Kang, B. B., Choi, H., Lee, H., and Cho, K.-J., 2019, "Exo-Glove Poly II: A Polymer-Based Soft Wearable Robot for the Hand with a Tendon-Driven Actuation System," *Soft Robotics*, **6**(2), pp. 214–227. <https://doi.org/10.1089/soro.2018.0006>.

[3] Oliver-Butler, K., Till, J., and Rucker, C., 2019, "Continuum Robot Stiffness Under External Loads and Prescribed Tendon Displacements," *IEEE Trans. Robot.*, **35**(2), pp. 403–419. <https://doi.org/10.1109/TRO.2018.2885923>.

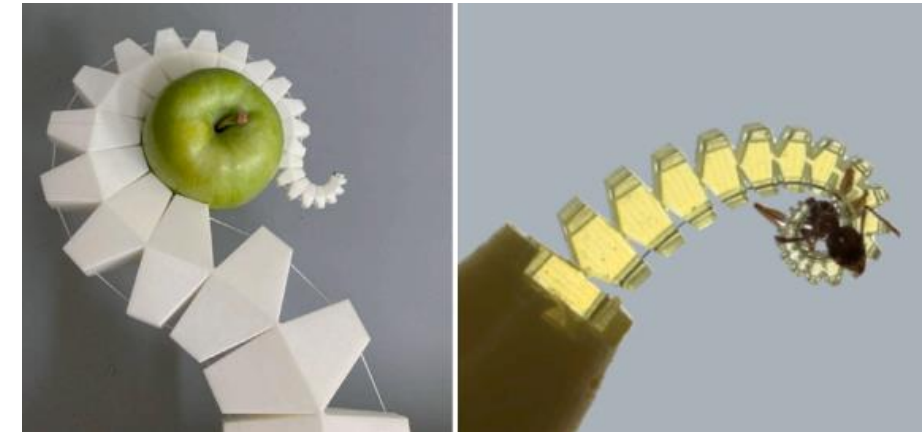
# Sequential Actuation



[4]



[3]



[5]

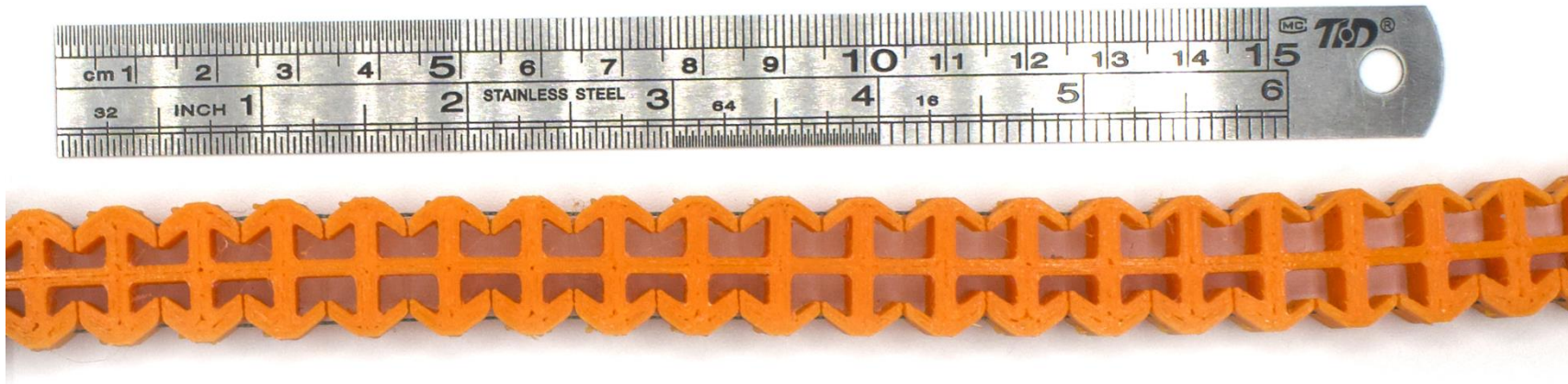
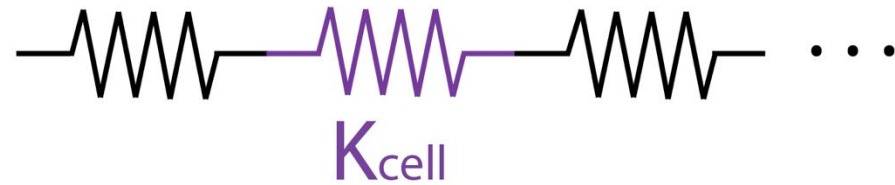
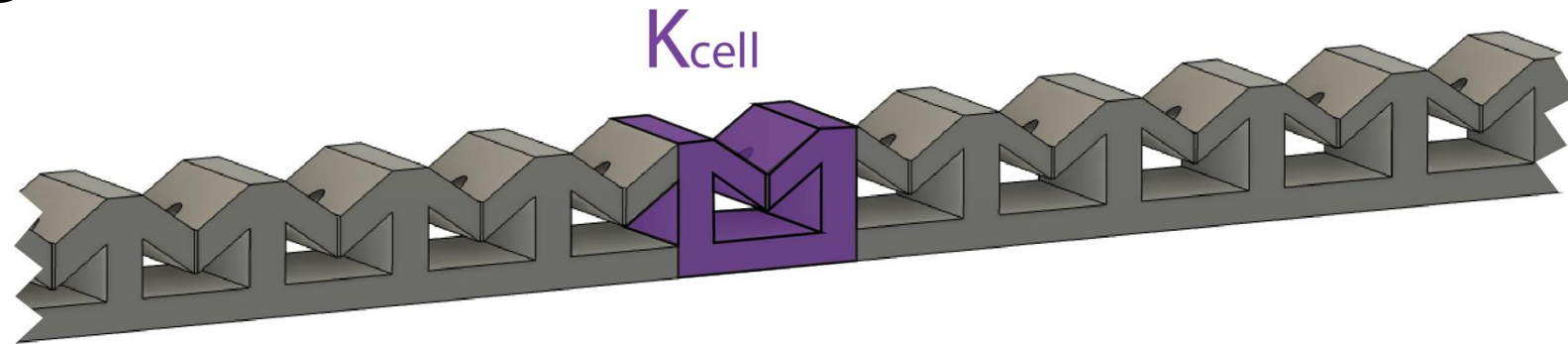
[4] Durham, J., "Curvature Computation for Sequentially Actuated Soft Graspers," MIT Department of Mechanical Engineering, Bachelor of Science Thesis, May 2025.

[5] Wang, Z., Freris, N. M., & Wei, X. (2025). SpiRobs: Logarithmic spiral-shaped robots for versatile grasping across scales. *Device*, 3(4).

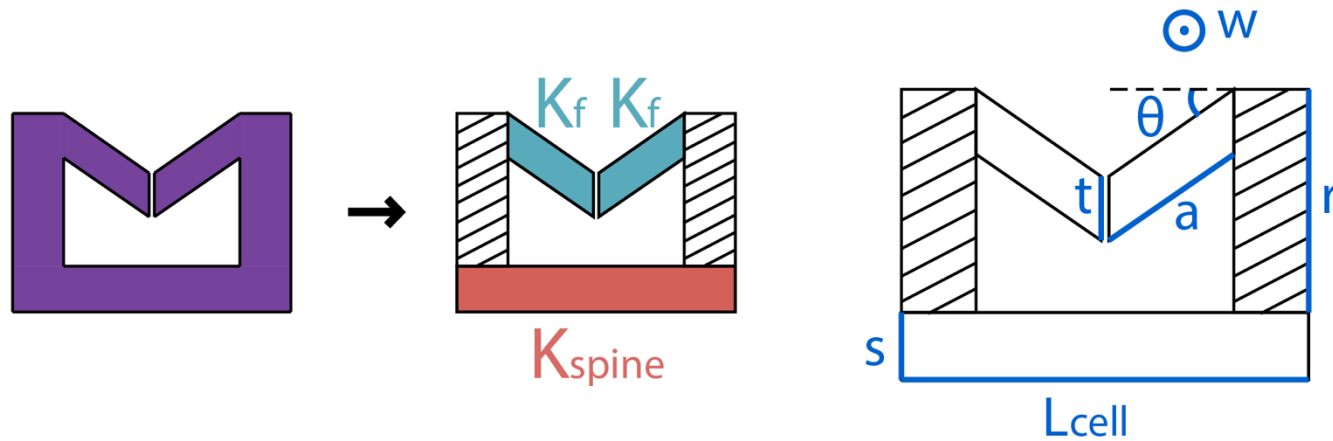
# The Design Problem

How can we develop a tendon-driven grasper with **programmable stiffness**, **is easy to fabricate**, and can grasp a **wide range of objects**?

# Design Process



# Design Process: PRBM



$$K_{cell} = \frac{K_f r^2}{2} + K_{spine}$$

$$K_{cell} = \frac{\gamma k_\theta E}{a^3 \cos(\theta) \sin(\theta)} * \frac{wt^3}{12} * \frac{r^2}{2} + \frac{Ews^3}{12L_{cell}}$$

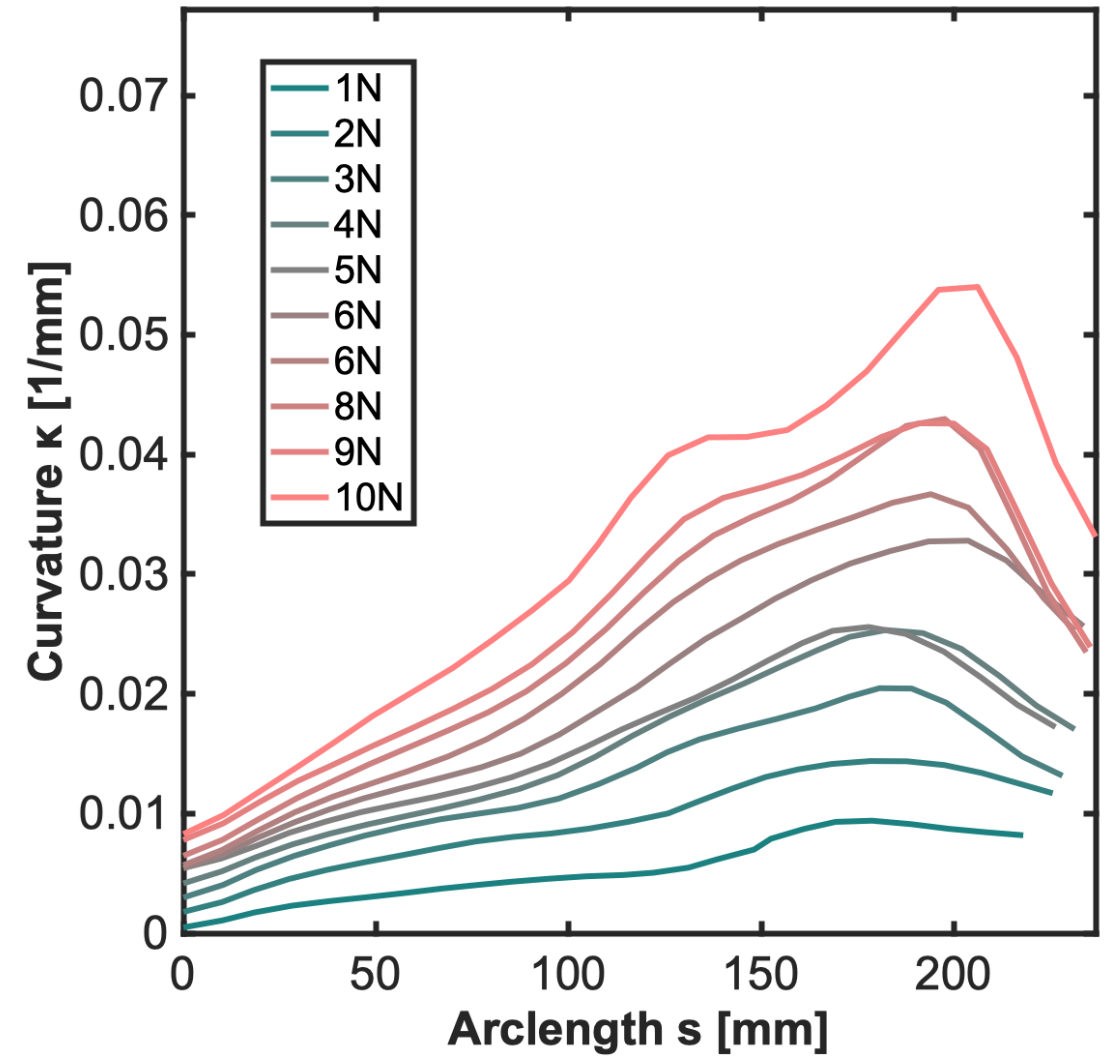
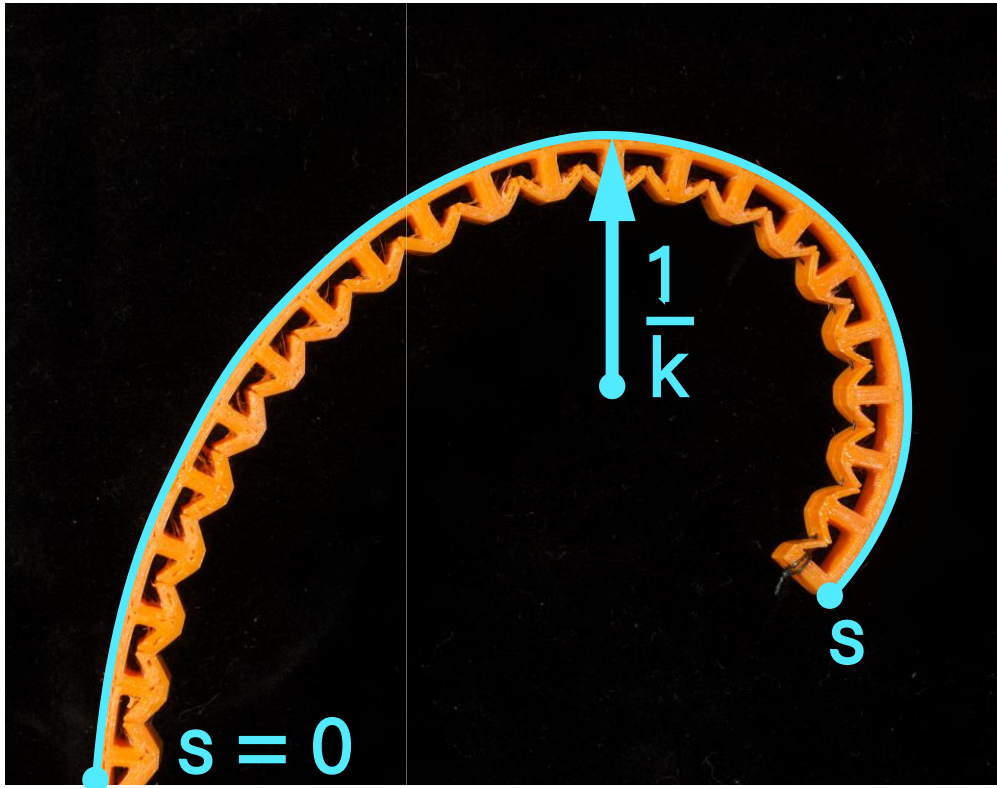
Pseudo-Rigid Body Model

$$K_f = \frac{\gamma k_\theta E}{a^3 \cos(\theta) \sin(\theta)} * \frac{wt^3}{12}$$

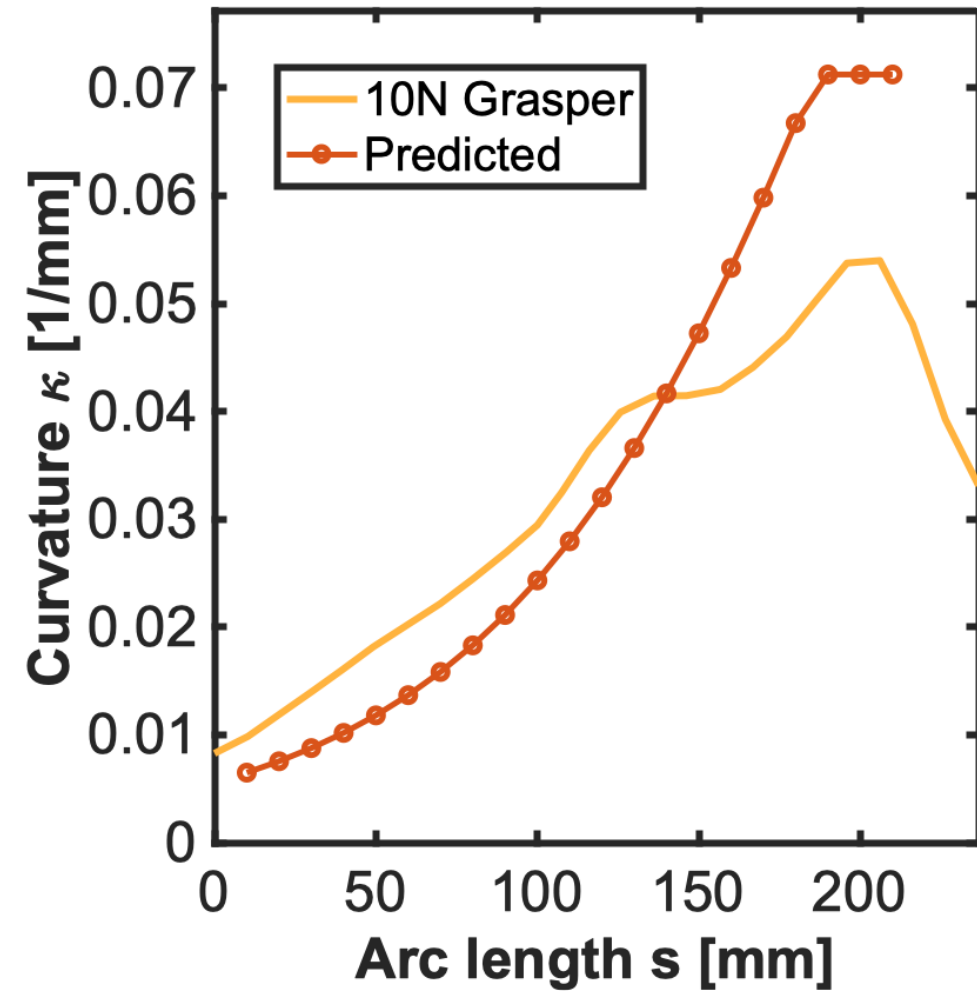
$$K_{spine} = \frac{Ews^3}{12L_{cell}}$$

$$\left. \begin{array}{l} \gamma \approx 0.85 \\ k_\theta \approx 2.5 \end{array} \right\} \text{PRBM constants}$$

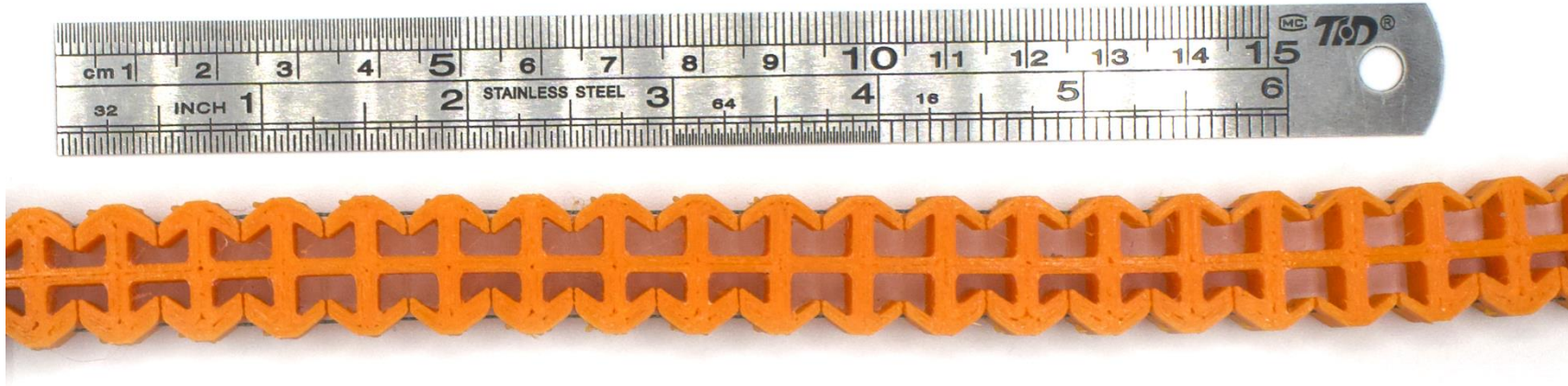
# Characterization



# Comparison to Model

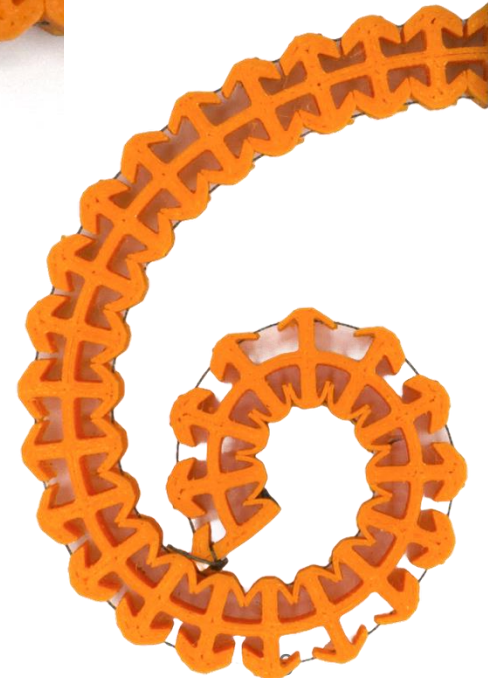


# Application of this design: Asymmetric Profile

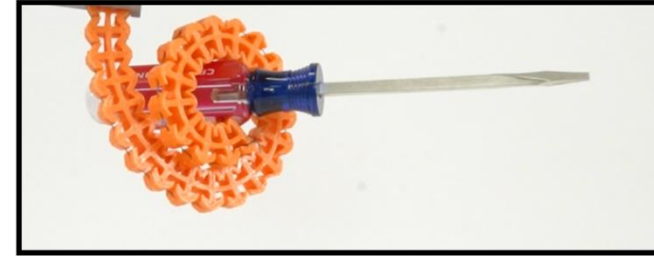


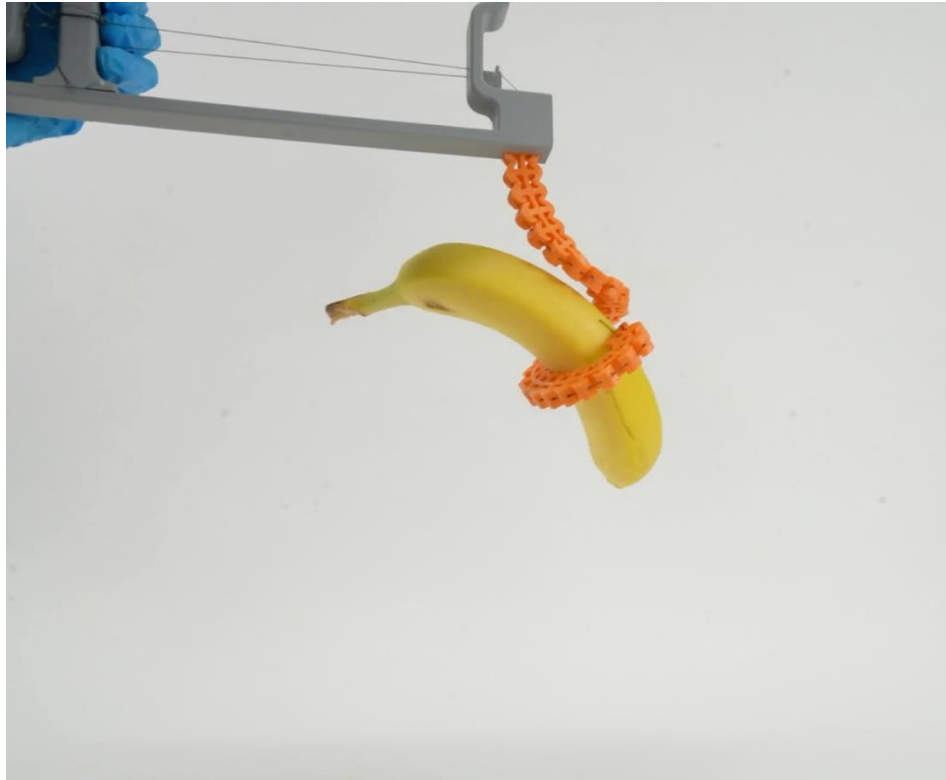
“Phalangeal”  
Actuation

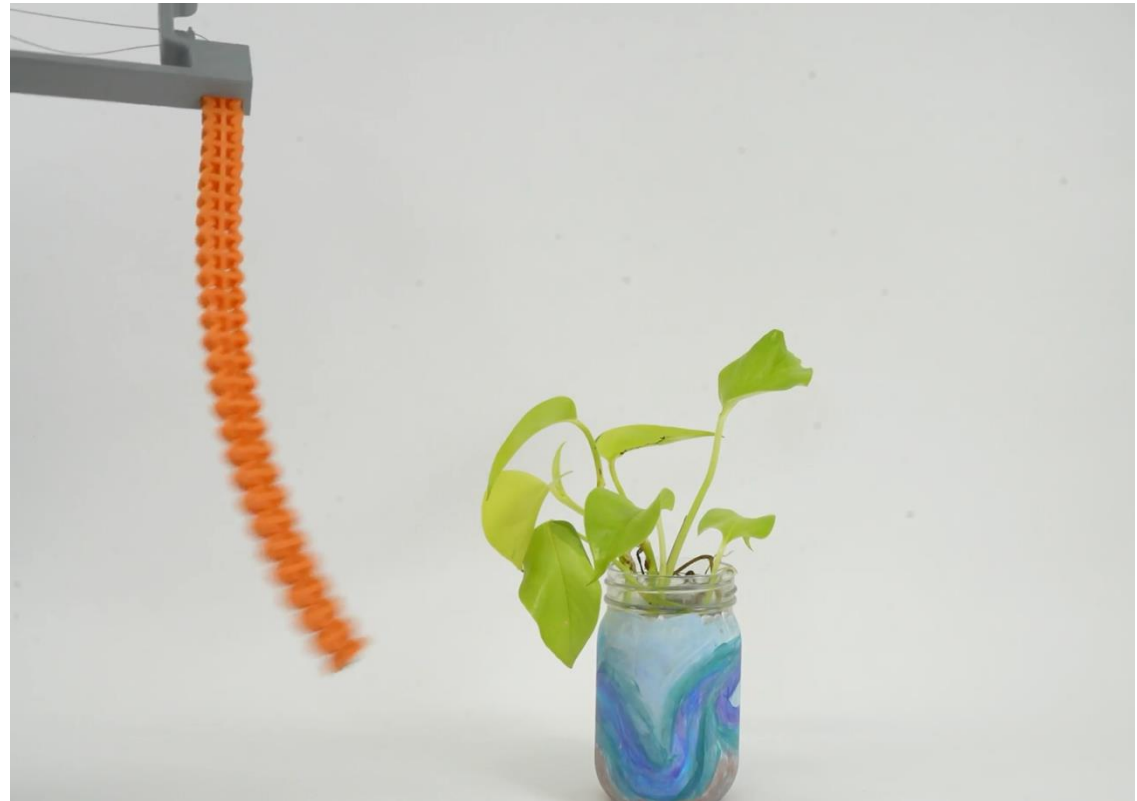
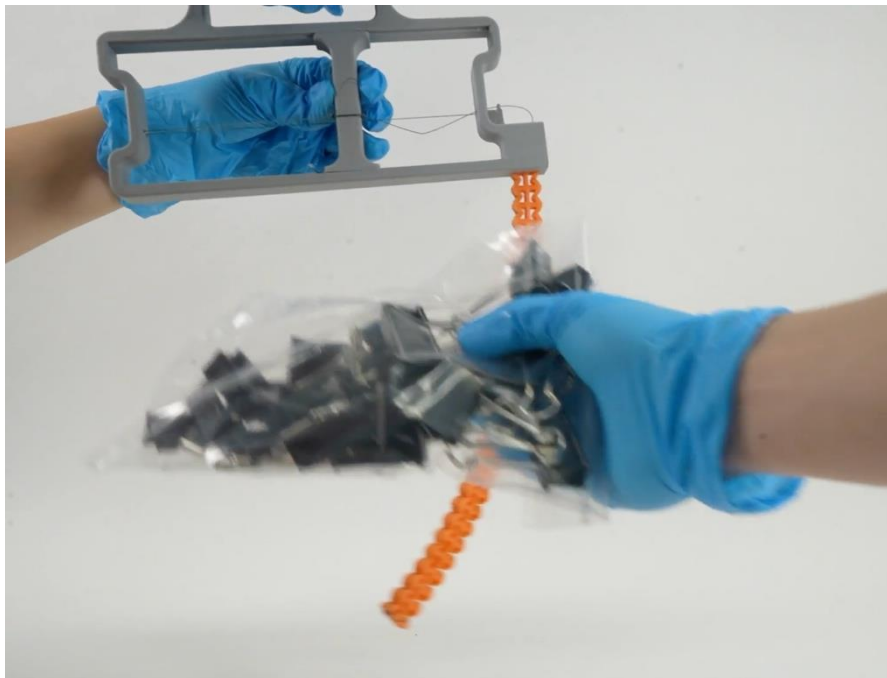
“Sequential”  
Actuation



# Demonstrations







# Conclusion

We achieved two distinct curling profiles without the need to reconfigure, enabling programmable and independent stiffness profiles and expanding grasping capability.

# Tendon-Driven Soft Grasper with Asymmetric Actuation through a Jamming Mechanism

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